

# Shaun Allison

shallison@protonmail.com  
<https://shalliso.github.io/>

## Education and Employment

Sessional Lecturer I, University of Toronto, [September 1, 2025 - December 31, 2025]

Postdoctoral fellow, University of Toronto, [August 1, 2022 - July 31, 2025]

Postdoctoral fellow, Hebrew University of Jerusalem, [July 1, 2021 to June 30, 2022].

Ph.D. Mathematics, Carnegie Mellon University, May 2021. 4.07 GPA

Dissertation: Classification of equivalence relations by Polish group actions

Advisor: Clinton Conley

B.S. Computer Science, Carnegie Mellon University, May 2015. 3.97 GPA

Thesis: A secure human-computable authentication scheme from the k-Junta problem

Advisors: Manuel Blum and Jeremiah Blocki

B.S. Mathematics (Discrete Mathematics and Logic), Carnegie Mellon University, May 2015. 3.97 GPA

## Other Employment

Full-time software engineer at Meta, June 2015- August 2016.

Software engineering internship at Meta, Summer 2014.

Research and development internship at VMware, Summer 2013.

Internship at Cisco, Summer 2012.

## Programming Languages and Technologies

I am currently using and have some experience in Lean and Python. I have some recent experience in statistical and machine learning tools in Python (numpy, matplotlib, pandas, pytorch). I use Latex regularly.

I have substantial previous but not recent experience with Hack, Javascript, Haskell, C, git

## Publications and Preprints

1. *The class and dynamics of  $\alpha$ -balanced Polish groups*, with Aristotelis Panagiotopoulos

Accepted **Journal of the European Mathematical Society**, 2026

Available from <https://arxiv.org/abs/2406.06082>

2. *Countable treeable Borel equivalence relations are classifiable by  $\ell_1$*   
**Advances in Mathematics**, 2025  
 Available from <https://arxiv.org/abs/2305.01049>
3. *Classification strength of Polish groups: Involving  $S_\infty$*   
 Accepted **Journal of Mathematical Logic**, 2026  
 Available from <https://arxiv.org/abs/2304.00139>
4. *Actions of tame abelian product groups*, with Assaf Shani  
**Journal of Mathematical Logic**, 2023  
 Available from <https://arxiv.org/abs/2105.05144>.
5. *Dynamical obstructions for classification by (co)homology and other TSI-group invariants*, with Aristotelis Panagiotopoulos  
**Transactions of the American Mathematical Society**, 2021  
 Available from <https://arxiv.org/abs/2004.07409>.
6. *Non-Archimedean TSI Polish groups and their potential Borel complexity spectrum*  
 Submitted 2020  
 Available from <https://arxiv.org/abs/2010.05085>.

## Talks

- Polish automorphism groups that don't involve  $S_\infty$* , University of Vienna, Set Theory Seminar (May 7, 2026)
- Revisiting Knight's model and its generalizations*, University of Vienna, Logic Colloquium (April 30, 2026)
- A formalization of a construction due to Julia Knight in the Lean 4 theorem prover with Mathlib*, Cornell, Logic Seminar (March 20, 2026)
- A formalization of a construction due to Julia Knight in the Lean 4 theorem prover with Mathlib*, Fields Institute, Set Theory Seminar (March 27, 2026)
- The algebra of Knight models*, Fields Institute, Set Theory Seminar (January 23, 2026)
- The Class of CLI Polish Groups is Coanalytic-Complete*, North American Descriptive Set Theory Meeting, Brin Mathematics Research Center (October 8, 2025)
- Automorphisms with Knight rank and bounds on cardinalities of potential Scott sentences*, Cornell Logic Seminar (April 24, 2025)
- Automorphisms with Knight rank and bounds on cardinalities of potential Scott sentences*, University of Maryland Logic Seminar (April 22, 2025)
- Automorphisms with Knight rank and bounds on cardinalities of potential Scott sentences*, CMU Logic Seminar (April 8, 2025)
- Ulm-like invariants for "almost smooth" Polish group actions*, Fields Institute, Set Theory Seminar (February 14, 2025)
- Treeable CBERs are classifiable by an abelian Polish group*, Fields Institute, Set Theory Seminar (November 22, 2024)

- A hierarchy of cli Polish groups and non-reducibility results*, Caltech logic seminar (November 6, 2024)
- A rank determining when a Polish automorphism group has less-than-maximal classification strength*, 2024 European Set Theory Conference (September 20, 2024)
- Lower bounds in potential Borel complexity, I and II*, Fields Institute, Set Theory Seminar (January 19 and 26, 2024)
- A hierarchy of cli Polish groups*, Fields Institute, Set Theory Seminar (November 10 and 17, 2024)
- Treeable CBERS are classifiable by an abelian Polish group*, Cornell Logic Seminar (November 3, 2023)
- Treeable CBERS are classifiable by an abelian Polish group*, Descriptive Dynamics and Combinatorics Seminar, Montréal (October 24, 2023)
- Polish groups involving  $S_\infty$* , Structures: Descriptive Set Theory and Dynamics, Institute of Mathematics of the Polish Academy of Sciences (August 15, 2023)
- Polish groups which don't involve  $S_\infty$* , 2023 North American Annual Meeting of the Association for Symbolic Logic, Descriptive Dynamics Special Session, University of California Irvine (March 27, 2023)
- Classification strength of Polish groups and involving  $S_\infty$* , Caltech logic seminar (January 31, 2023)
- Polish groups involving  $S_\infty$* , 2022 CMS Winter Meeting, Session on set theory and its applications (December 3, 2022)
- Classification strength of Polish groups I and II*, Fields Institute, Set Theory Seminar (October 7 and 14, 2022)
- Polish groups with the pinned property I and II*, Hebrew University, Set Theory Seminar (May 11 and 18, 2022)
- Applications of the Hjorth analysis of Polish group actions*, Hebrew University, Set Theory Seminar (December 22, 2021)
- Potential Borel complexity and tame abelian product groups*, Cornell Logic Seminar (March 26, 2021)
- Polish groups with the pinned property*, CMS 2020 Winter Meeting, Canadian Mathematical Society (December 8, 2020)
- Potential Borel complexity and tame abelian Polish groups*, Set Theory Reading Group, Carnegie Mellon University (November 24, 2020)
- An anticlassification result for TSI Polish groups*, Mathematical Logic Seminar, Carnegie Mellon University (November 24, 2020)
- Classification by TSI Polish groups*, Logic Seminar, University of Illinois at Chicago (September 1, 2020)
- Non-Archimedean TSI Polish groups and obstructions to Borel reducibility*, Logic Seminar, California Institute of Technology (February 12, 2020)
- An effective proof of the  $(Go, Ho)$  dichotomy I and II*, Mathematical Logic Seminar, Carnegie Mellon University (November 5, 2019 and November 12, 2019)
- Applications of the Gandy-Harrington topology*, Boise Extravaganza in Set Theory, AAAS Pacific Division Annual Meeting, (June 20, 2019)
- Generic preturbulence and forcing*, Mathematical Logic Seminar, Carnegie Mellon University (April 30, 2019)

*Hjorth isomorphism and orbit equivalence relations*, Mathematical Logic Seminar, Carnegie Mellon University (April 23, 2019)

*Model theory and the Glimm-Effros property*, Graduate Students Conference in Logic, University of Wisconsin-Madison (April 21, 2018)

## Teaching

At University of Toronto:

1. *Instructor of record*, MAT135 Differential Calculus (Fall 2025) approx. 400 students
2. *Instructor of record*, MAT405 Introduction to Topology (Winter 2025) approx. 19 students
3. *Instructor of record*, MAT309 Introduction to Logic (Fall 2024) approx. 16 students
4. *Instructor of record*, MAT401 Polynomial Equations and Fields (Winter 2023) approx. 15 students
5. *Instructor of record*, MAT309 Introduction to Logic (Fall 2023) approx. 11 students
6. *Instructor of record*, MAT244 Differential Equations (Winter 2023) approx. 110 students
7. *Instructor of record*, MAT102 Introduction to Proofs (Fall 2022) approx. 80 students

At Carnegie Mellon University:

1. *Instructor of record*, 21-127 Concepts of Mathematics (Summer 2018) approx. 20 students
2. *Teaching assistant*, Matrices and Linear Transformations (Fall 2020), Concepts of Mathematics (Fall 2019 and Fall 2016), Matrix Theory (Fall 2018), Calculus in Three Dimensions (Spring 2018 and Fall 2017), Calculus I (Spring 2017)
3. *Grader*, Linear Algebra (Spring 2020), and Set Theory (Spring 2020 and Spring 2019)

## Other Activities

Formalized the construction of Knight's model in Lean

Lead an independent study with an undergraduate on model theory during the Winter 2023 semester

Organized a reading seminar on the manuscript "Geometric Set Theory" by Paul Larson and Jindrich Zapletal, and related works, during the Fall 2019 semester

Created a new informal weekly seminar for graduate students in set theory and model theory at Carnegie Mellon University in 2019, still running as of 2023.

## Patents

I have two patents with a colleague during my time at Meta that involve machine learning.

Automatic recipient targeting for notifications (US10506055B2) with L. Foged (*Granted*)

Selecting assets (US20180027092A1) with L. Foged (*Granted*)